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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Minor Degree Examinations

December 2025

EC3108-1 – JAVA PROGRAMMING

(For Minor Degree)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following loops will always execute at least once?
A) for loop
B) while loop
C) do-while loop
D) Both while and for loop
2. In Java, an abstract class can have:
A) Only abstract methods
B) Only concrete methods
C) Both abstract and concrete methods
D) No methods
3. What happens if you explicitly call super() in a constructor?
A) Calls the parent class constructor
B) Skips the constructor
C) Throws a runtime exception
D) Calls the static method
4. Which of the following is NOT a primitive data type in Java?
A) int
B) float
C) boolean
D) string
5. What is the output of the following code?

```
for(int i = 0; i < 5; i++) {  
    System.out.print(i + " ");  
}
```


A) 0 1 2 3 4 5
B) 0 1 2 3 4
C) 0 1 2 3
D) Compilation Error
6. Which of the following correctly declares an abstract method?
A) void display() abstract;
B) abstract();
C) abstract display() void;
D) abstract void display();
7. Which of the following keywords is used to inherit a class in Java?
A) extends
B) implements
C) inherits
D) superclass
8. A method is
A) A variable within a class
B) A block of code that performs a specific task
C) A data type
D) A keyword
9. Which of the following is NOT a checked exception?
A) IllegalArgumentException
B) ClassNotFoundException
C) NullPointerException
D) FileNotFoundException
10. What is the purpose of the synchronized keyword in Java?
A) To allow multiple threads to access a method simultaneously
B) To pause a thread
C) To prevent thread interference
D) To stop a thread
11. What is the purpose of the finally block in exception handling?
A) To execute code after the catch block, regardless of exception
B) To define another exception type
C) To handle system errors
D) To rethrow an exception

12. Which of the following blocks cannot be skipped during exception handling?
 A) try B) catch
 C) throw D) finally
13. Which method is used to execute a SQL query in JDBC?
 A) executeStatement() B) runQuery()
 C) runStatement() D) executeQuery()
14. In Java's exception hierarchy, which class is the direct superclass of all exceptions?
 A) Throwable B) Exception
 C) RuntimeException D) Error
15. Which interface in JDBC is specifically designed to execute precompiled SQL queries with parameters?
 A) Statement B) PreparedStatement
 C) CallableStatement D) ResultSet
16. What type of exception is ArithmeticException?
 A) Checked exception B) Unchecked exception
 C) Error D) IOException
17. Which class is used to establish a connection with a database in JDBC?
 A) ConnectionFactory B) Statement
 C) ResultSet D) DriverManager
18. Which of these packages contains all the classes and methods required for event handling in Java?
 A) java.applet B) java.awt.event
 C) java.awt D) java.event
19. What role does an event listener play in the event handling model?
 A) It generates the event B) It dispatches the event to the appropriate method
 C) It listens for and handles events D) It queues the events
20. What exception is commonly thrown when an attempt to open a file fails in Java?
 A) IOException B) FileNotFoundException
 C) EOFException D) NullPointerException

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the loop control Statements available in Java with suitable examples.	8	L*2	1	1
	b) Explain multilevel inheritance in Java with a suitable program.	8	L2	2	1
2.	a) Using an example program, explain the basic structure of a Java program.	8	L1	1	1
	b) How can run-time polymorphism be achieved through dynamic method dispatch in Java? Explain with an example.	8	L2	2	1
3.	a) Briefly explain Type promotion and Type conversion with separate JAVA codes.	8	L2	1	1
	b) Create a superclass named point2D with instance variables x and y, a parameterized constructor to initialize the instance variables. Create a subclass named point3D with instance variables x, y, and z. Demonstrate single inheritance for the above-mentioned specifications. Invoke super superclass constructor from the subclass.	8	L3	2	1

Unit – II

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|-------|--|---|----|---|---|
| 4. a) | Explain in detail the life cycle of a thread in Java with suitable block diagram. | 8 | L2 | 3 | 1 |
| b) | Explain the JDBC Architecture and its components. How does JDBC facilitate database connectivity in Java applications? | 8 | L2 | 4 | 1 |
| 5. a) | How are exceptions handled in JAVA? Explain briefly with an example program. | 8 | L2 | 3 | 1 |
| b) | What is a JDBC driver? Explain four different types of JDBC drivers in detail. | 8 | L2 | 4 | 1 |
| 6. a) | How to create a Thread by implementing a Runnable interface? Illustrate with the help of a program. | 8 | L2 | 3 | 1 |
| b) | Differentiate between Statement, PreparedStatement, and CallableStatement in JDBC with suitable examples. | 8 | L2 | 4 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | What are the key methods provided by the Java File class for performing file operations? Explain each method with its use. | 8 | L2 | 5 | 1 |
| b) | Define an event listener in Java. How does it work and what is its role in event-driven programming? | 8 | L2 | 5 | 1 |
| 8. a) | How does Java handle events in GUI applications? Describe the event handling mechanism with explanation | 8 | L2 | 5 | 1 |
| b) | How do you create a file in Java? Illustrate with the help of sample code. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
